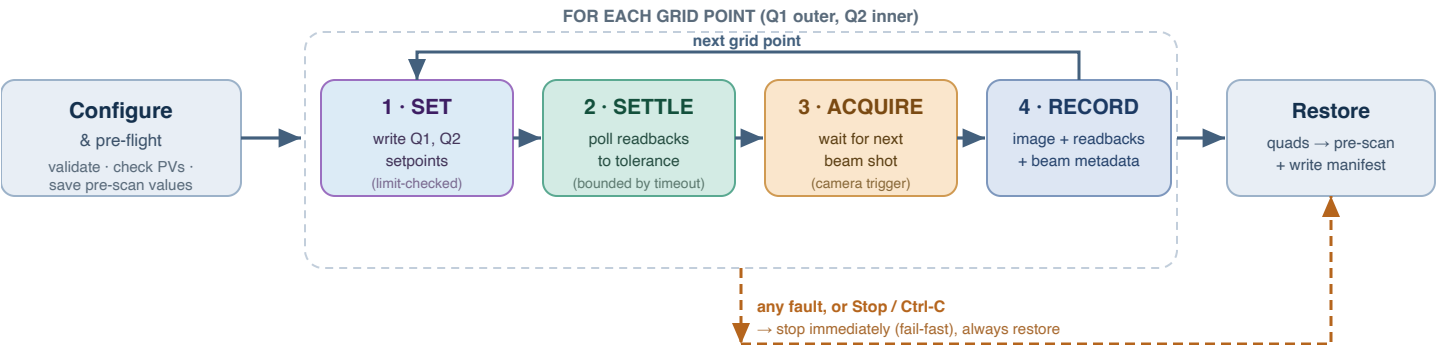


Automated 2D Quadrupole Scan

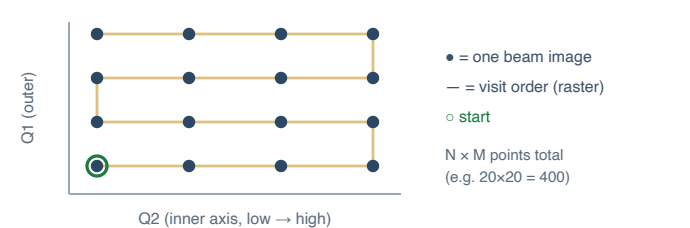
Beam-spot imaging system · quadrupole-scan data acquisition for emittance measurement

The system records a beam-spot image on the YAG screen at each point of a two-dimensional grid of quadrupole settings. Two quadrupoles (**Q1**, **Q2**) are stepped across a configured range; at every setting the magnets are allowed to settle and the next beam shot is captured. The result is a complete, labeled image set — one frame per (Q1, Q2) pair — together with the magnet readbacks and beam parameters recorded at the moment of each shot, which is the raw material a quadrupole-scan emittance analysis consumes.

HOW A SCAN RUNS



SCAN GRID & ORDER



- **Raster, unidirectional:** Q1 holds while Q2 sweeps low→high, then Q2 resets and Q1 advances.
- Each axis is set by **min**, **max**, and number of **points**, inside a fixed hard-limit band.
- **Frames per point** is configurable (default 1); multiple shots per setting are all kept as raw frames.

BEAM DELIVERY

- The accelerator timing system and operator **fire the beam**; the program does **not** trigger the gun.
- The camera is **hardware-triggered** by the shot, so each captured frame is a real beam shot.
- “Acquire” waits for the next shot **after** the magnets settle; shots arriving mid-move are discarded. Shot cadence ≈ 1.3 s.

WHAT GETS RECORDED

One run folder per scan, containing for every frame:

- **Raw image** — 16-bit TIFF of the beam spot.
- **Sidcar file** — Q1/Q2 setpoints & readbacks, timestamps, beam parameters.
- **manifest.csv** — one row per frame (the full index of the run).
- **scan_config.json** — exact parameters the run used.

Beam parameters logged per shot: target bucket, gun bunches, injection mode, gun-inhibit state, sequence number (from the injection request) and gun-bias readback — so each image is traceable to the beam that made it.

SAFETY & INTEGRITY

- **Limit-checked writes:** every magnet setpoint is refused if it falls outside that quad’s configured safe band — checked both before the scan and at every write.
- **Pre-flight:** the scan refuses to start unless the whole grid is within limits and all required signals are connected.
- **Fail-fast + restore:** any fault (a magnet not settling, no beam within the timeout, a lost signal) stops the scan at once and drives both quads back to their pre-scan values.
- **Stop any time:** a manual stop follows the same safe restore path; partial data is preserved.

SCOPE

The scan produces the labeled image set and metadata for a quad-scan measurement. **Spot-size fitting and emittance computation are performed separately**, downstream from this acquisition.

Acquisition runs from either a command-line driver or a “Scan” panel in the bench GUI; both share the same scan logic, safety checks, and output format. Estimated duration ≈ (number of grid points) × (settle time + one beam cycle); e.g. a 20×20 grid is on the order of 10–20 minutes depending on magnet settling.